Digital Sentinels: AI-Driven Risk Signal Detection for Substance Abuse

Blake Simpson, Rohan Ashraf Hashmi, Kenneth Kakie

1. Executive Summary and Project Goals

Substance abuse remains a critical public health crisis, often characterized by a lag between community-level behavioral shifts and official medical reporting. Early warning signals frequently emerge in digital environments, such as online communities and medical review forums, before manifesting in clinical data.

The primary objective of this project was to design an AI-driven system capable of identifying risk signals related to alcohol and drug abuse from public, anonymized data (specifically the UCI Drug Review dataset). The overarching mandate was to transform unstructured, noisy text into interpretable, actionable insights that support early public health awareness and intervention.

This mandate was divided into two core goals:

1. **Risk Signal Detection:** Developing a robust, scalable NLP pipeline to extract relevant textual signals (substance mentions, emotional distress, relapse indicators) while navigating nuanced language and slang.
2. **Temporal and Behavioral Analysis:** Modeling the temporal dynamics of these signals to detect anomalies (spikes) and clustering users to uncover evolving behavioral narratives and early-warning indicators.

2. System Architecture and Methodology

To achieve the project goals, a modular, pipeline-based architecture was developed. The system was designed to prioritize computational efficiency, data integrity, and analytical depth. The methodology was codified into specific programmatic modules.

2.1. Data Contracts (`schemas.py`)

- **Purpose:** Defines rigid Pydantic models (`RawReview`, `ProcessedReview`, `DetectionResult`) to structure the data at every stage of the pipeline.
- **Logic and Significance:** Real-world datasets are inherently messy, often containing missing values or malformed rows. By enforcing data contracts, the system guarantees that the downstream temporal and behavioral analysis engines receive consistent,

validated data. This prevents pipeline failures during batch processing and standardizes the output for reporting.

2.2. Ingestion and Memory Management (`Loader.py`)

- **Purpose:** Utilizes pandas to load the dataset, normalize datetime formats, and stream the data in manageable chunks via a generator function.
- **Logic and Significance:** Processing hundreds of thousands of text records simultaneously can cause catastrophic memory overflows, especially when generating high-dimensional vectors. The batch-streaming approach ensures the system remains lightweight and can scale to much larger datasets without requiring enterprise-grade hardware.

2.3. Text Normalization (`preprocessor.py`)

- **Purpose:** Implements a `FastPreprocessor` utilizing compiled regular expressions to strip HTML artifacts, special characters, and redundant whitespace.
- **Logic and Significance:** Speed was prioritized over exhaustive natural language processing (NLP). By bypassing heavy libraries like `spaCy` in favor of compiled regex, the system rapidly sanitizes text to ensure the downstream embedding models are not confused by web-scraping artifacts.

2.4. Hybrid Detection Engine (`dictionaries.py` & Engine Logic)

- **Purpose:** Combines a Rule-Based Baseline (fast regex matching against specialized clinical and slang vocabularies defined in `dictionaries.py`) with a Semantic Embedding Model (`all-MiniLM-L6-v2`) to detect distress.
- **Logic and Significance:** This hybrid approach balances precision and recall. Hardcoded dictionaries perfectly capture explicit mentions of controlled substances or relapse keywords. The local embedding model acts as a semantic safety net, catching veiled or nuanced expressions of psychological distress that lack specific keywords. Running embeddings locally avoids the latency, cost, and privacy concerns of external LLM APIs.

2.5. Behavioral Clustering and Validation (`processing_results.ipynb`)

- **Purpose:** Utilizes K-Means clustering on the generated embeddings to group similar reviews, extracts topics via TF-IDF, and validates the clusters using the Silhouette Score and test-set generalization.

- **Logic and Significance:** This module transforms mathematical vectors back into human-interpretable themes. By calculating the Silhouette Score, the system statistically proves that the narratives are distinct, non-random behavioral patterns. Testing these clusters against unseen data ensures the model has learned universal abuse narratives rather than memorizing historical artifacts.

3. Results: Evolving Narratives and Risk Profiles

The system successfully processed the text and clustered the high-risk signals into distinct behavioral narratives. Cross-referencing the cluster assignments with the drug distribution heatmap revealed profound structural patterns in how different substances correlate with specific types of distress.

Cluster 0: Psychological Distress & Mental Health

- **Keywords:** *anxiety, withdrawal, suicidal, depression, thoughts*
- **Prominent Drugs:** Medications prescribed for neurodivergence, insomnia, panic, anxiety, and depression (e.g., Benzodiazepines, SSRIs, Z-drugs).
- **Interpretation:** This cluster captures the severe psychiatric toll of dependency, tolerance, or rapid cessation of neuro-active medications. The presence of suicidal ideation alongside withdrawal highlights a critical intervention vector for mental health professionals.

Cluster 1: Physical Dependency & Pain Management

- **Keywords:** *pain, withdrawal, months, years, effects*
- **Prominent Drugs:** Opioids, heavy painkillers, and euphoric/discomfort-reducing substances (e.g., Oxycodone, Fentanyl, Tramadol).
- **Interpretation:** This is the classic signature of chronic pain patients struggling with physiological opioid dependency. The emphasis on time ("months," "years") suggests long-term users facing diminishing returns from their prescriptions and struggling with physical withdrawal symptoms.

Cluster 2: The Semantic False Positive (Physical Side Effects)

- **Keywords:** *weight, lost, cravings, eat, started*
- **Prominent Drugs:** Weight-loss medications, stimulants, and specific antidepressants.

- **Interpretation:** This cluster highlights a collision in semantic sentiment. The AI successfully identified "cravings" as a high-risk flag; however, contextually, these users are discussing *food* cravings and physical side effects, not chemical dependency. While not directly showing abuse patterns, this successfully isolated dataset noise into a distinct bucket, keeping the other clusters pure.

Cluster 3: Medication-Assisted Treatment (MAT) & Recovery

- **Keywords:** *suboxone, methadone, day, years, life*
- **Prominent Drugs:** Anti-opioid and maintenance drugs.
- **Interpretation:** This narrative shifts away from active abuse toward recovery and maintenance. The vocabulary here ("life," "years") often indicates users documenting their sobriety journeys or discussing the difficulties of tapering off MAT drugs.

4. Limitations and Potential Improvements

While the pipeline successfully generated actionable insights, the results highlighted specific areas for technical refinement.

4.1. The HTML Entity Leakage (')

- **Issue:** The TF-IDF keyword extraction prominently featured 039, which is the numerical remnant of the HTML entity for an apostrophe (').
- **Improvement:** The `FastPreprocessor` must be updated. By importing Python's built-in html module and running `html.unescape(text)` before applying the special-character-stripping regex, these artifacts will be cleanly converted to standard punctuation and subsequently removed.

4.2. Resolving the Semantic False Positive (Cluster 2)

- **Issue:** The system overly penalized the word "cravings," capturing dietary side effects rather than substance abuse.
- **Improvement:** Implement a **Context-Aware Weighting system** within the `RuleDetector`. By using a small token window (e.g., ±3 words), the system can demote the risk score if the word "craving" appears adjacent to "food," "sugar," "eat," or "weight." Alternatively, creating a rigid "Substance of Interest" (SoI) filter at the ingestion stage would drop weight-loss drugs from the pipeline entirely.

4.3. Dynamic Thresholding for Spikes

- **Improvement:** The current 2σ Z-score threshold for spike detection is static. Implementing an exponentially weighted moving average (EWMA) would make the system more sensitive to sudden, acute crises, reducing the lag in early-warning detection.

5. Future Extensions and Scalability

To transition this project from a static analysis tool to a live, operational public health instrument, several extensions should be pursued.

5.1. Real-Time Social Media Ingestion

- **The Vision:** Transitioning from static CSVs to daily API polling from platforms like Reddit (e.g., r/Opiates, r/recovery), Twitter/X, and specialized web forums.
- **The Value:** Daily pulling shifts the system into a true "Early Warning" posture. It allows for the detection of zero-day threats, such as the sudden emergence of a new synthetic street drug or a contaminated batch of narcotics in a specific geographic locale, days or weeks before emergency room data reflects the crisis.

5.2. Multi-Modal Data Overlay

- **The Vision:** Integrating the digital signal data with hard public health metrics, such as the CDC Overdose Data or NIDA statistics.
- **The Value:** By running a cross-correlation analysis, the system could prove that a "Cluster 1 Spike" in digital domains reliably predicts a localized surge in hospitalizations 14 days later. This transitions the AI from a descriptive tool to a predictive epidemiological model.

5.3. Targeted LLM Triage

- **The Vision:** While the current system avoids LLMs for batch processing speed, an LLM (like Gemini or GPT-4) could be deployed surgically.
- **The Value:** When the pipeline detects a statistical "Risk Spike," the system could automatically package the top 50 most representative reviews from that spike and send them to an LLM. The LLM would then generate an automated, human-readable "Crisis Briefing" (e.g., *"Users are currently reporting extreme withdrawal symptoms due to a*

sudden shortage of generic Oxycodone at major pharmacy chains"), instantly providing policymakers with the exact context of the anomaly.